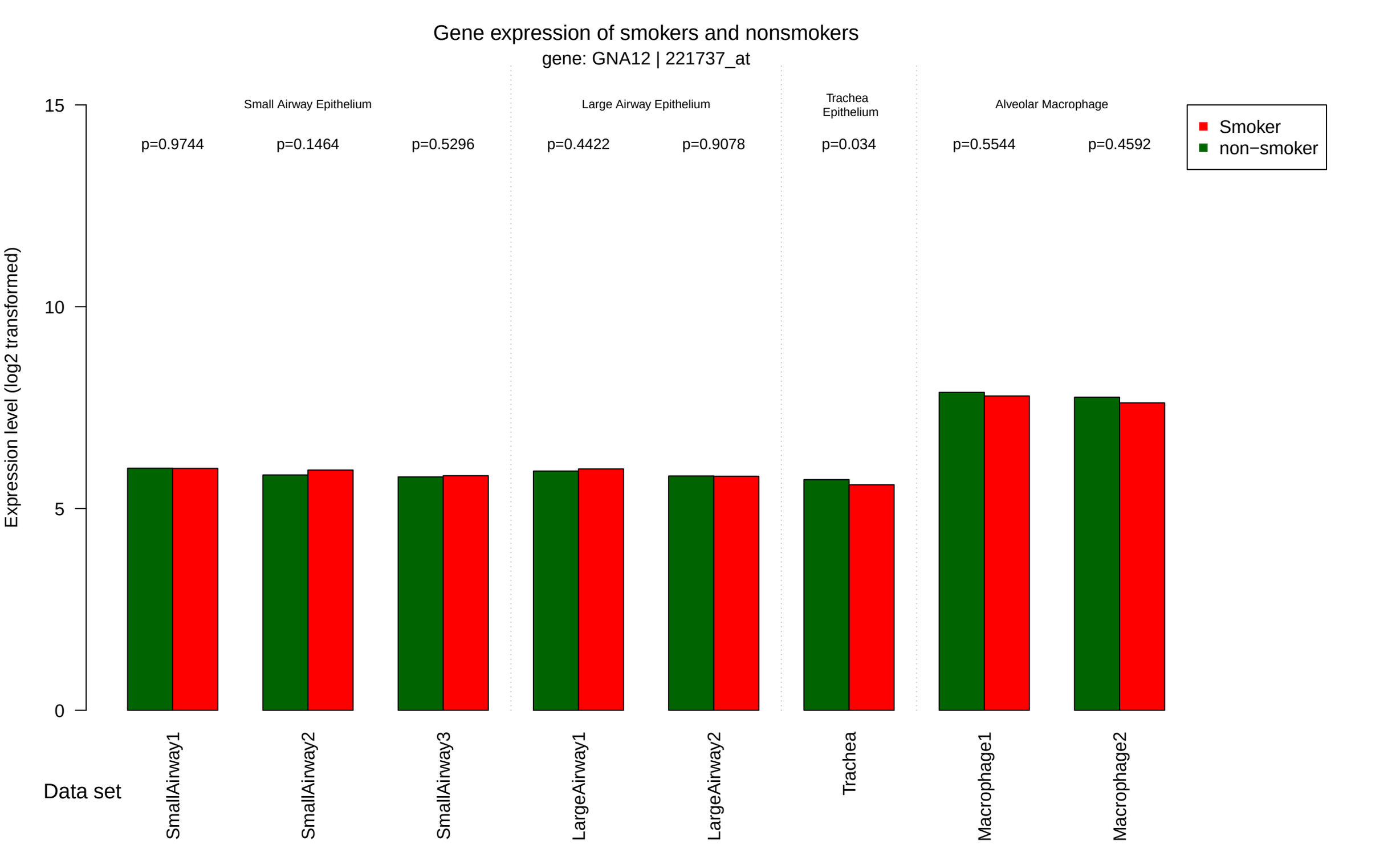
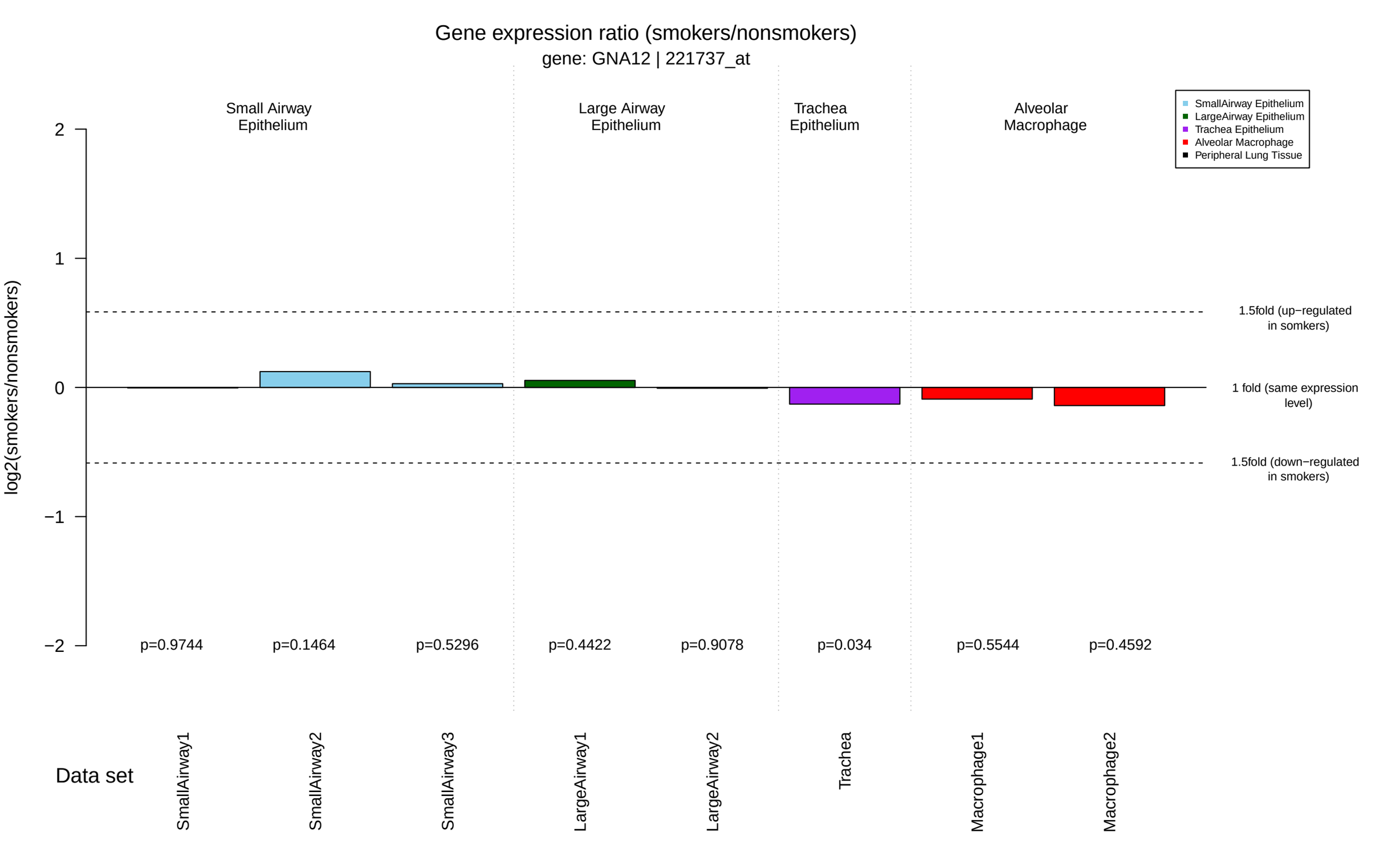
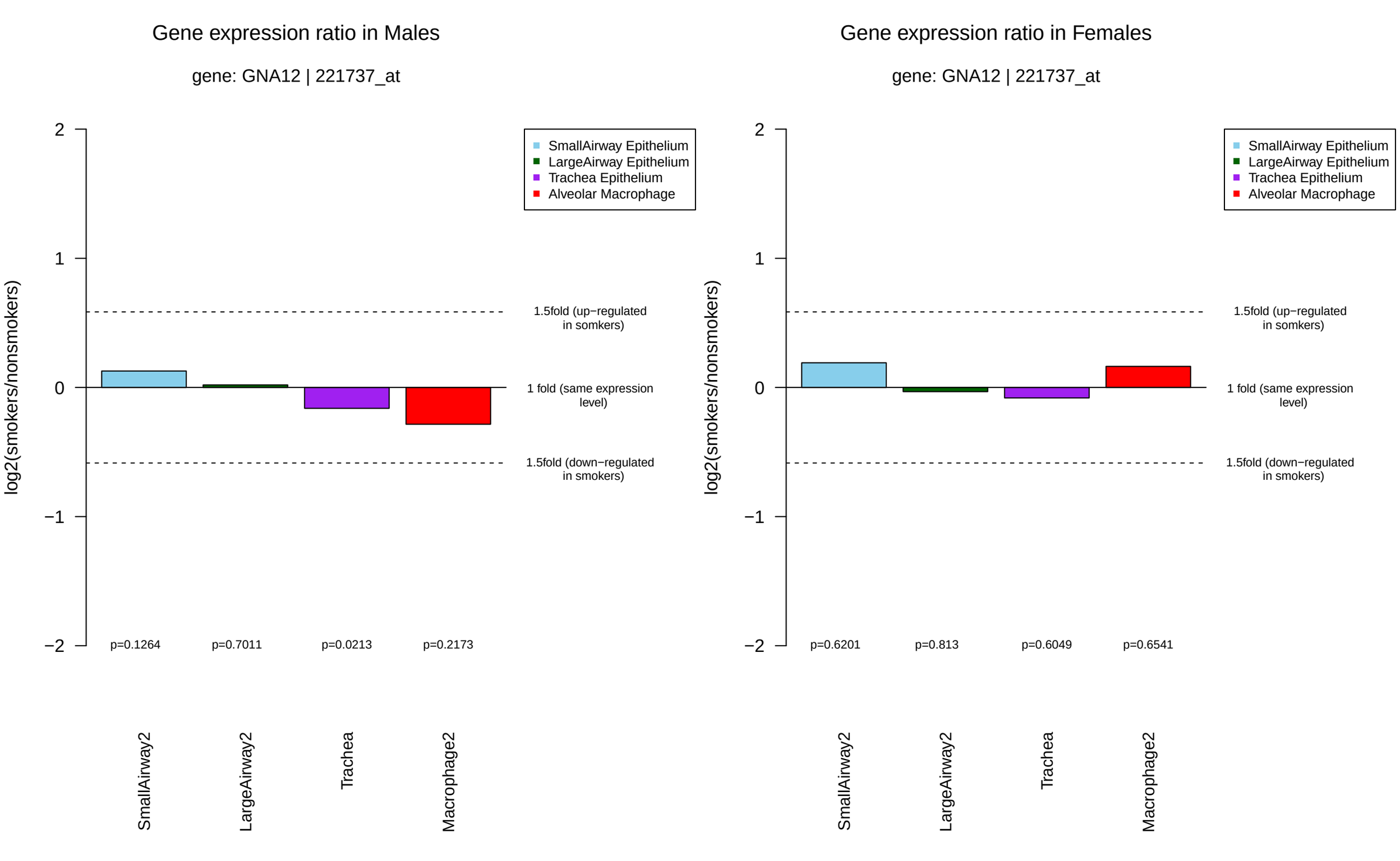
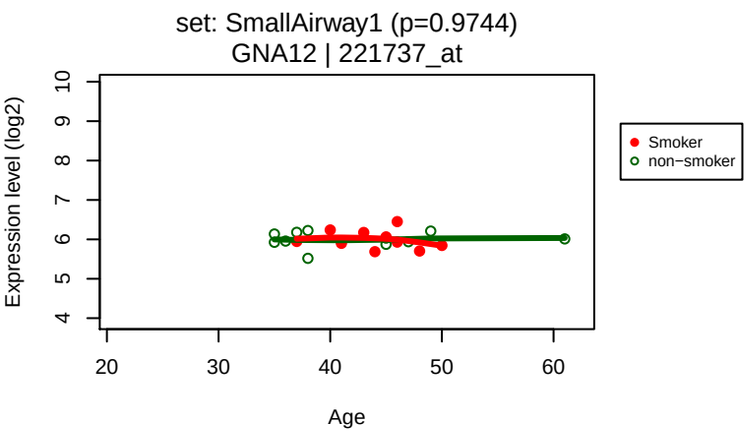
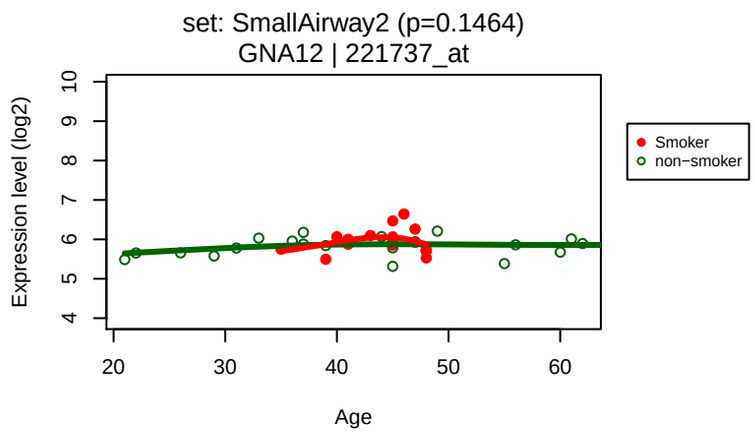
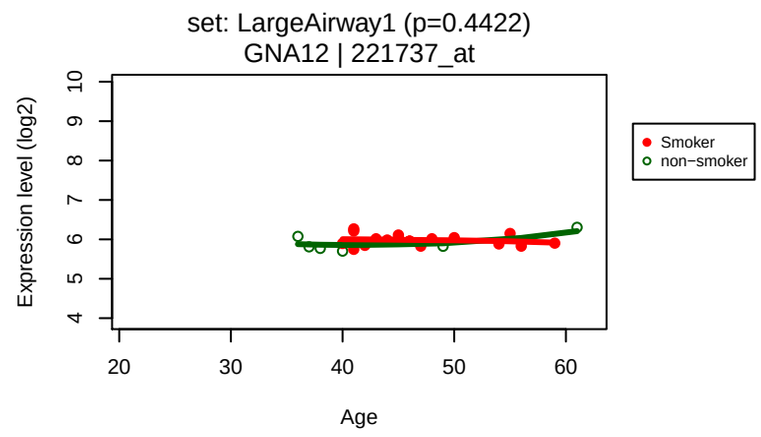

Gene expression of smokers and nonsmokers
gene: GNA12 | 221737_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.9744
p=0.1464
p=0.5296
p=0.4422
p=0.9078
p=0.034
p=0.5544
p=0.4592
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

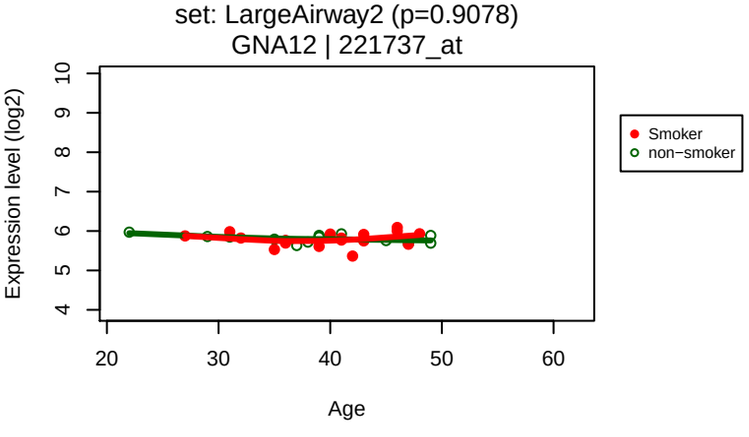

Gene expression ratio (smokers/nonsmokers)
gene: GNA12 | 221737_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9744
p=0.1464
p=0.5296
p=0.4422
p=0.9078
p=0.034
p=0.5544
p=0.4592
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

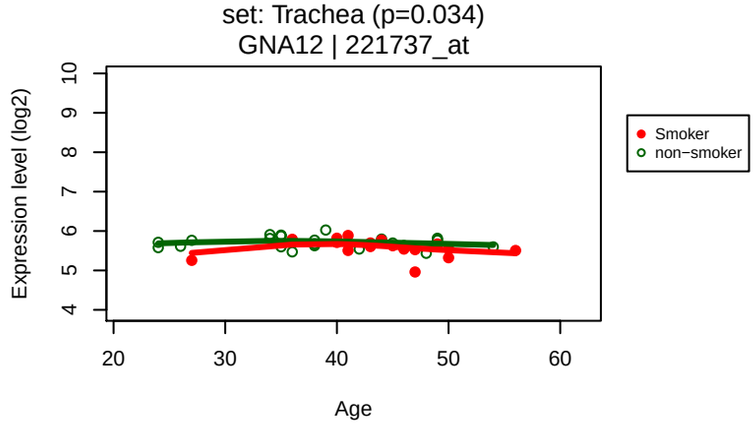

Gene expression ratio in Males
gene: GNA12 | 221737_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1264
p=0.7011
p=0.0213
p=0.2173
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: GNA12 | 221737_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6201
p=0.813
p=0.6049
p=0.6541
SmallAirway2
LargeAirway2
Trachea
Macrophage2

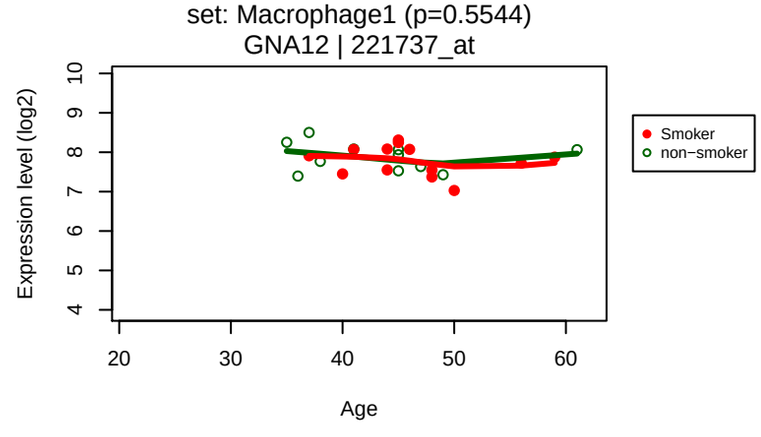

set: SmallAirway1 (p=0.9744)
GNA12 | 221737_at
Expression level (log2)
Age
Smoker
non−smoker

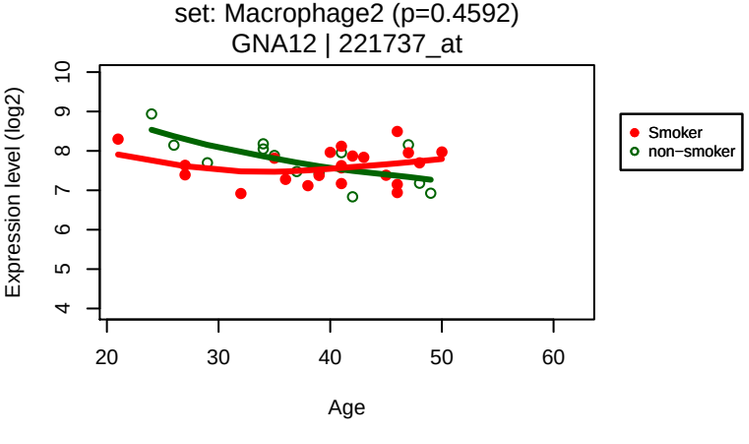

set: SmallAirway2 (p=0.1464)
GNA12 | 221737_at
Expression level (log2)
Age
Smoker
non−smoker

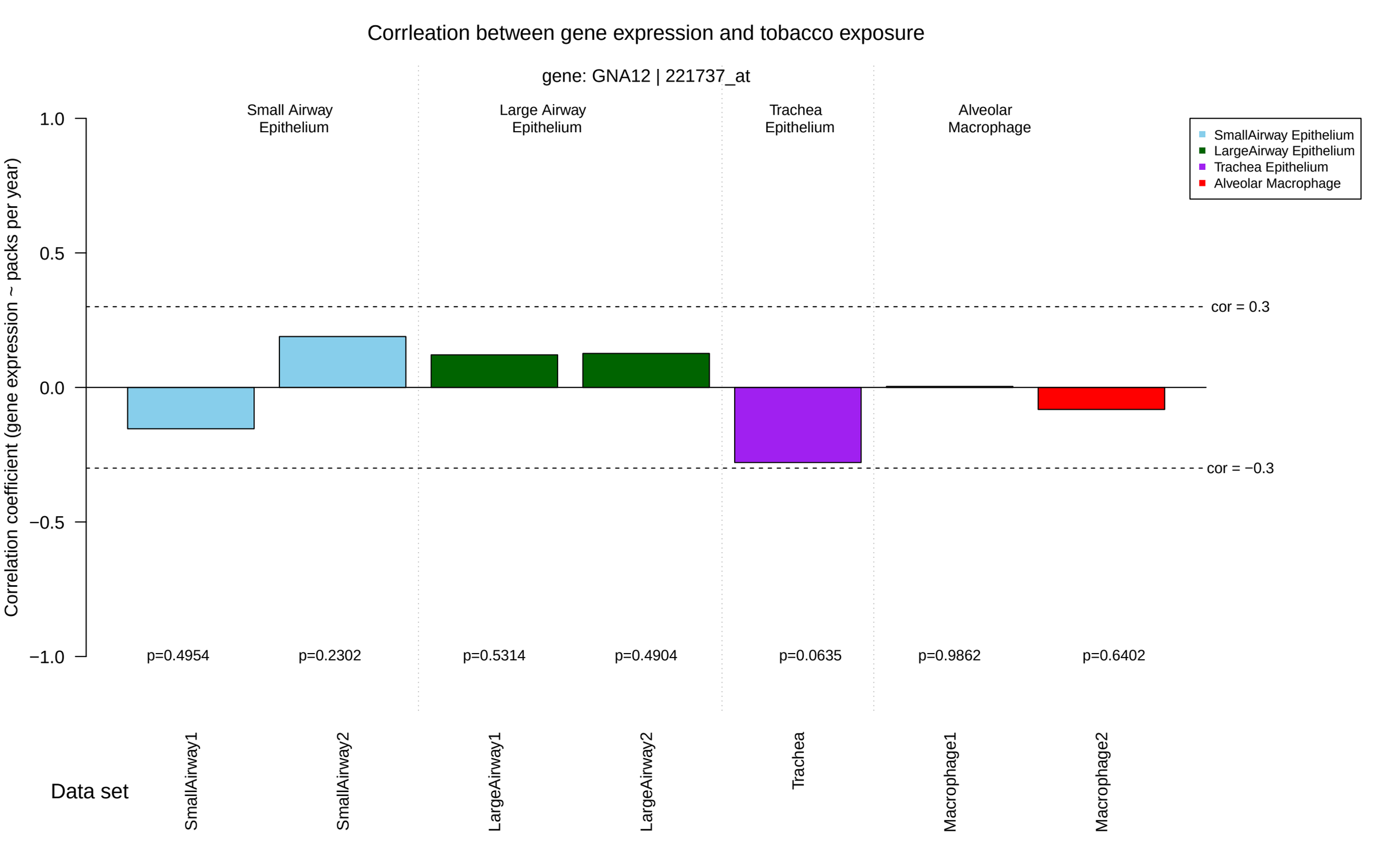

set: LargeAirway1 (p=0.4422)
GNA12 | 221737_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.9078)
GNA12 | 221737_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.034)
GNA12 | 221737_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.5544)
GNA12 | 221737_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.4592)
GNA12 | 221737_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: GNA12 | 221737_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.4954
p=0.2302
p=0.5314
p=0.4904
p=0.0635
p=0.9862
p=0.6402
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2